

Alex Turianskyj

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🎓 EDUCATION

McGill University, B.Eng. in Software Engineering Co-op

GPA: 3.69/4.00 | Aug 2021 – Dec 2025

💻 SKILLS

Programming: TypeScript, Java, Python, C#, C++, SQL
Tools: Git, GitHub Actions, Docker, AWS, Linux, Bash
Testing: Cypress, Jest, JUnit, Pytest, Catch2, Selenium

Frontend: React, Angular (NgRx), Tailwind, HTML, CSS
Backend: Spring Boot, .NET, FastAPI, PostgreSQL
Languages: English (native), French (native)

💼 WORK EXPERIENCE

Autodesk

Software Developer, Platform Services (Contract) | Spring Boot, AWS Montreal, QC | Feb 2026 – Present

- Designed and implemented v3 Asset Graph REST endpoints in **Spring Boot** using **OpenSpec** for spec-driven development, translating request schemas to a gRPC backend for bulk reads and filtered search queries.
- Maintained a federated **GraphQL** supergraph, resolving 5+ critical security vulnerabilities and executing a zero-downtime Apollo Federation upgrade across 3 subgraphs using **GraphOS** and **Spinnaker**.
- Built multi-region **Catchpoint** synthetic checks via parameterized Kubernetes ConfigMaps, integrating Slack alerts and PagerDuty escalation to detect production outages and retire legacy internal monitoring.
- Accelerated client migration to v3 APIs by modernizing an internal **Claude** plugin for onboarding and integration.

Software Developer Intern, Fusion | React, TypeScript, C++ Montreal, QC | May 2025 – Aug 2025

- Stabilized the **React** Properties panel for real-time collaborative editing in Fusion (200,000+ customers) by eliminating 5+ UI regressions and increasing **TypeScript** test coverage using **Jest**.
- Resolved critical caching failures by refactoring **C++** JSON parsing to decouple model and component events.
- Expanded deep links in the Fusion client to open past versions of a design, enhancing web-to-desktop workflows.
- Reduced incomplete panel analytics payloads by **80%** by extending serialization for booleans and nested objects.
- Authored foundational **Cypress** tests and integrated LaunchDarkly flags for a controlled panel layout rollout.

Matrox, Software Engineering Intern | Angular, ASP.NET Core, C# Montreal, QC | Jan 2024 – Aug 2024

- Developed a **TypeScript** monitoring dashboard with live charting of device metrics to streamline remote analysis.
- Built **Angular** components from Figma designs with **NgRx** state management and extended the **ASP.NET Core** REST API for full-stack functionality, enabling key device features including volume and keyboard layout control.
- Engineered a **C#** logging provider to transmit device logs via **SignalR**, accelerating root-cause troubleshooting.

Hydro-Québec, Software Development Intern | JavaScript, VBA Montreal, QC | May 2023 – Aug 2023

- Engineered a custom **JavaScript** testing tool and **VBA** automation scripts, reducing data validation time by **95%+**.

🔗 SELECTED PROJECTS

Vibe 📄, Python, FastAPI, SvelteKit, NumPy, Polars, Docker Jan 2026 – Mar 2026

- Developed a content-based music recommender with **4M+** songs and **200k+** artists, deployed at **vibe.alex.t.dev** 📄
- Designed a ranking engine using soft-min aggregation across **12** audio features, **80+** genres, and **50+** languages.
- Optimized backend via candidate pre-filtering and **Polars** data structures to maintain **<1GB** RAM on a 2-core VPS.

Holoportation 📄, C#, C++, Python, .NET, OpenCV Sep 2024 – Apr 2025

- Built an AR system for HoloLens 2 astronaut training, named a **Top 5 Finalist** for the McGill Engine Capstone Prize.
- Enhanced LiveScan3D in **C++** and **C#** with WinForms for real-time 3D reconstruction using Femto Bolt cameras.
- Integrated **OpenCV** and YOLO-World to extract documents from depth-masked RGB streams for AR readability.

Daily Ball 📄, Unity, C#, Android, WebGL May 2023 – Jul 2024

- Developed a physics-based 2D mobile game in **Unity** and launched it on Google Play, achieving **1,000+** downloads.
- Increased player retention by **30%** through game updates and released a WebGL demo at **dailyball.alex.t.dev** 📄